Lesson plan · for the teacher

What makes a choice the best?

Optimization · Level 1 · Lesson 1

Time	Materials
40 min	Board and chalk; Five cards per group, each with an outing (or a class reward), its cost and its votes; A budget on the board, such as 1,200 coins

Learning goals

- Name the options, the limits and the goal of a decision.
- Tell a plan that keeps every limit from the best plan.
- Find the best plan for our class outings, and ask who it leaves out.

Activity: The class outings

1. Warm-up (5 min): Ask pupils what would make the best day out. Collect answers (cheap, fun, close, everyone together) and point out that people mean different things by “best”.
2. Teach (8 min): Write three headings on the board: options (what we choose), limits (what must hold) and goal (what “best” means). Fill them in for the outings: five ideas, a budget of 1,200 coins, the most votes.
3. Vote (5 min): Write five outing ideas with their costs, such as zoo 700, museum 500, farm 400, river park 300 and science fair 200. Each pupil votes for two; write the totals beside each idea.
4. Find the best (15 min): Groups use one card per idea and try mixes, adding up the cost of each. A plan over budget is not allowed. Each group writes its best plan and its votes on the board.
5. Wrap-up (7 min): Compare the plans. Did “favourite first” win? Which plan has the most votes? Ask whose favourite was left out and agree one fair rule for next term, such as taking turns.

Prepare one set of five cards per group: the outing, its cost and space to write its votes. No outings planned? Use class rewards instead (a story hour, a football match, a film).

Discussion

- Why didn’t “favourite first” give the most votes?
- How do we know a plan is the best, and not just allowed?
- Whose favourite was left out, and what would be fair next time?

Exit ticket

- Write the options, the limits and the goal of one decision at our school.

Try it in class tomorrow

Plan the class outings together: write five ideas with their costs, let pupils vote, and set a budget. Groups find the mix with the most votes inside the budget and explain why it’s the best. Then ask: whose favourite was left out, and what would be fair next term?

Worksheet · for pupils

What makes a choice the best?

Optimization · Level 1 · Lesson 1

Name _____ Class _____ Date _____

1 **A colleague says: "The best plan for the class outings always starts with the class favourite, the zoo." Is she right?**

◯ Yes: the favourite belongs in every best plan ◯ No: a mix of smaller choices can score more

◯ Only when the favourite is the cheapest

2 **The school kitchen plans lunch. Match each sentence to the part of the decision it describes.**

How many portions of rice, beans and greens to serve	• • • •	A feasible plan
Every pupil gets at least 5 mg of iron	• • • •	The goal
Spend as little money as possible	• • • •	A limit
A plate that meets every need, whatever it costs	• • • •	The options

3 **Adding a new limit, such as "every class gets at least one outing", can never make the best plan score higher on the goal.**

◯ True ◯ False

4 **Which of these are limits for a school timetable? Choose all that apply.**

☐ A teacher can't teach two classes at once ☐ Ms Huda leaves after period 3

☐ As few gaps as possible in teachers' days ☐ Each class has maths four times a week

☐ Which period each lesson goes in

5 **Complete the sentence.**

A plan that keeps every _____ is _____; the one with the best score on the goal is _____.

Word bank: limit · feasible · optimal · goal · option

✎ **Think of a decision people argue about at your school: the timetable, duties, a trip. What are its options, limits and goal, and do people agree on the goal?**